Inputs

File: Inputs.kicad_sch

AFE

File: AFE.kicad_sch

Digital

File: Digital.kicad_sch

Power

File: Power.kicad_sch

12-lead portable ECG — 10 electrodes, 8 simultaneous 24-bit channels

45 x 45 mm, 4 layers; protected 1S cell; USB-C charging; BLE batches <=1 s target

Read DESIGN_SPEC.md and DESIGN_REPORT.md before prototype assembly.

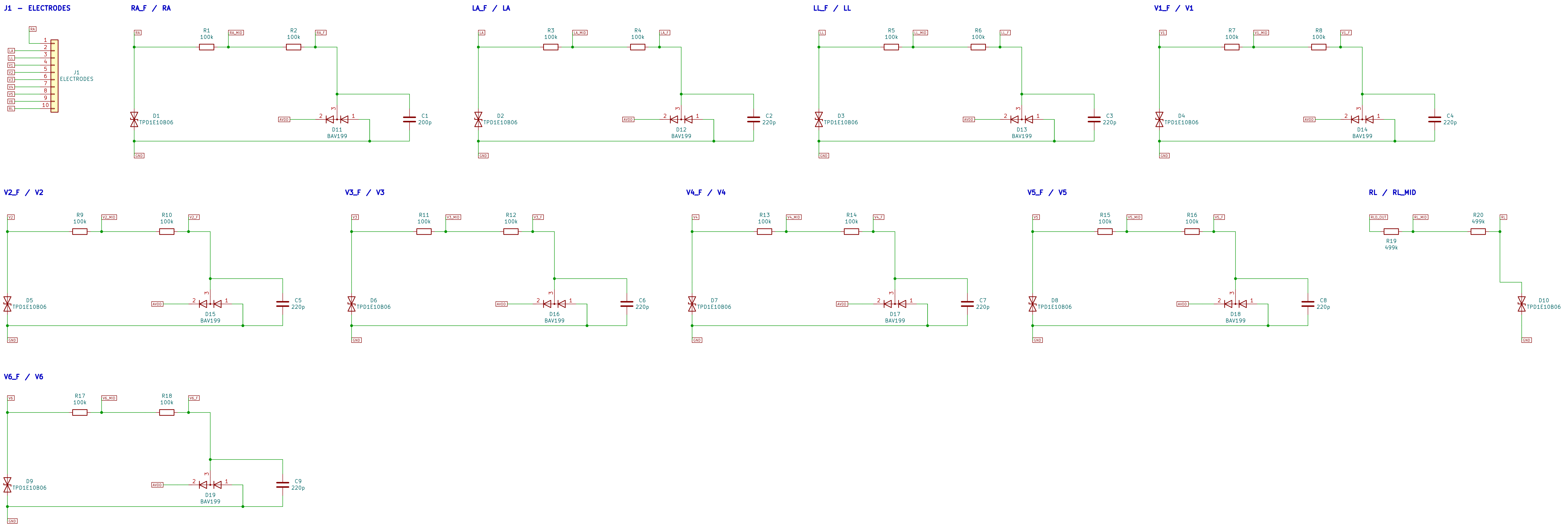
Only connect electrodes when running from battery; USB/programming requires electrode removal.

ELECTRODE PROTECTION / MATCHED RF FILTERS

RA/LA/LL and V1–V6: TVS –> 2x100k –> low–leakage clamp –> COG –> AFE.

All COG capacitors 50 V. Match total input capacitance during CMRR verification.

RL: 2x499k series output limit. No defibrillator or electrosurgical protection.



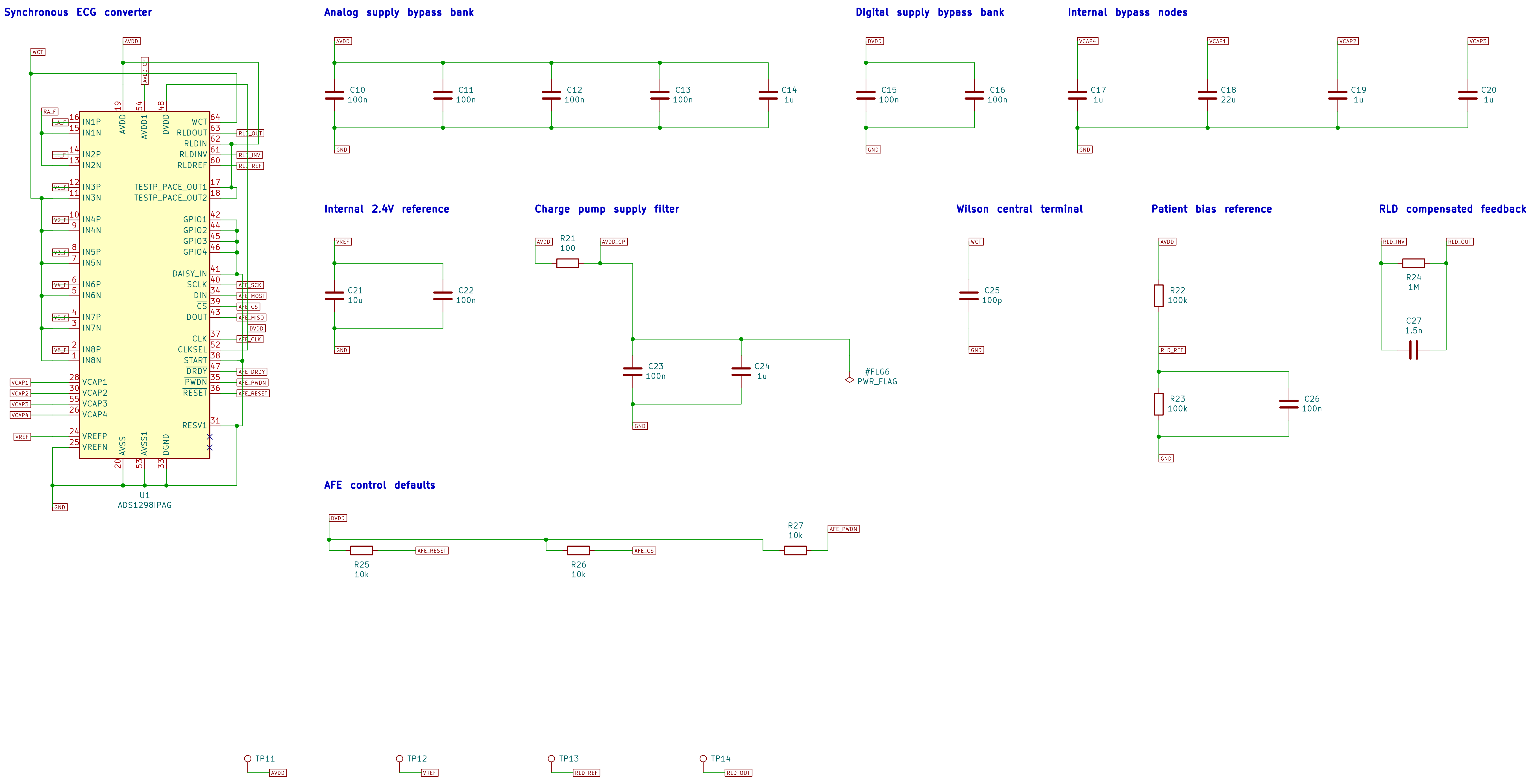
8–CHANNEL SYNCHRONOUS AFE / RLD / WILSON TERMINAL

CH1=LA–RA, CH2=LL–RA; CH3–8=V1–V6 minus WCT.

WCT1=0x09, WCT2=0xC2: RA=CH1N, LA=CH1P, LL=CH2P.

START pin low: use SPI START/STOP commands. Keep unused GPIO inputs.

Internal VREF=2.4 V, GAIN=6; 500 SPS monitoring (sinc response is not flat to 150 Hz).



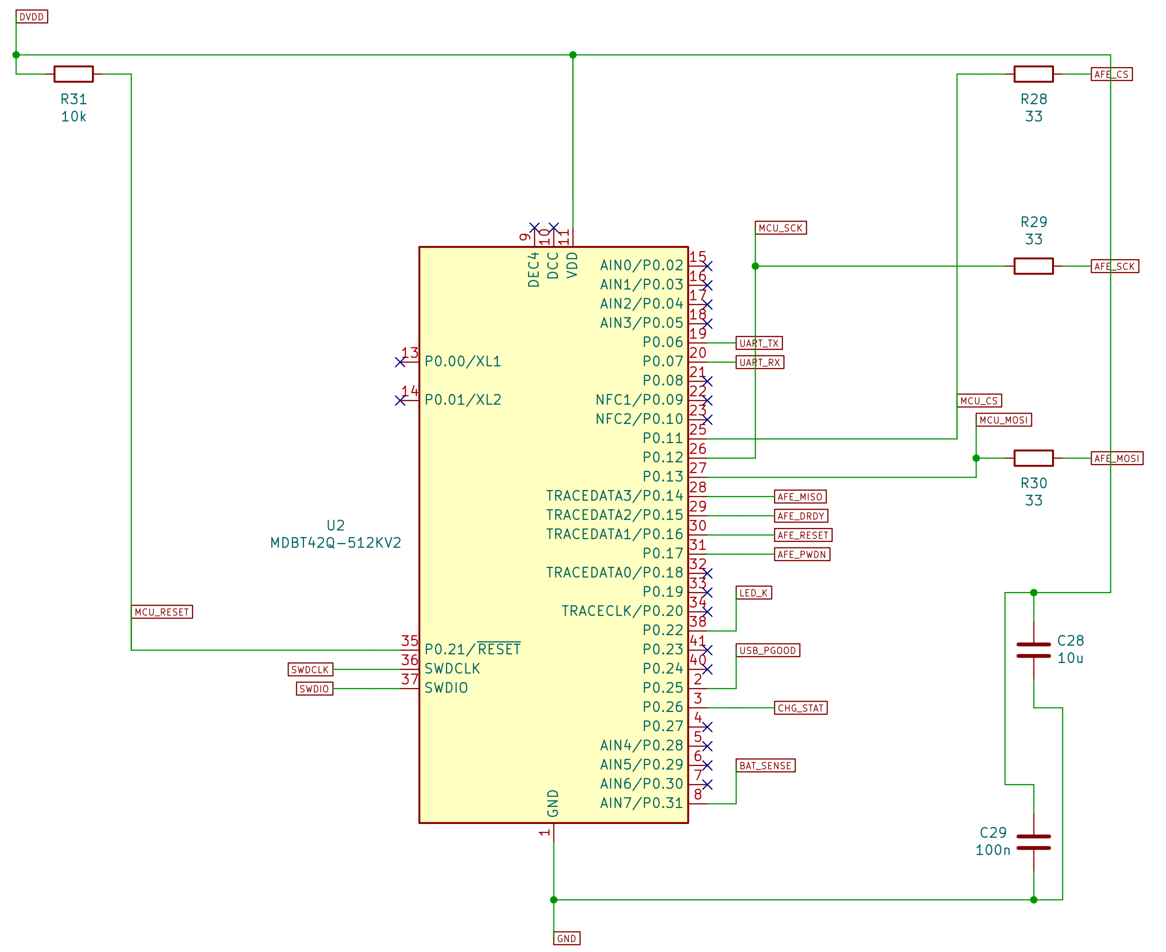
BLE OPEN–CPU MODULE / BUFFERING / TEST

Firmware: LDO mode, calibrated LFRC; no external DCC/DEC4 network.

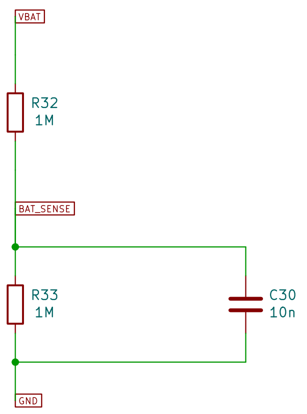
32 KiB RAM ring, batch every 250 ms; measured payload target >=200 kbit/s.

Maximum 1 s data–age target requires compatible central and bounded radio retries.

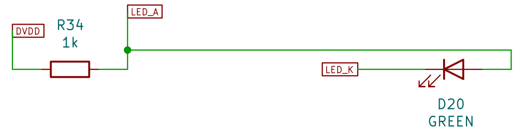
BLE module and SPI termination



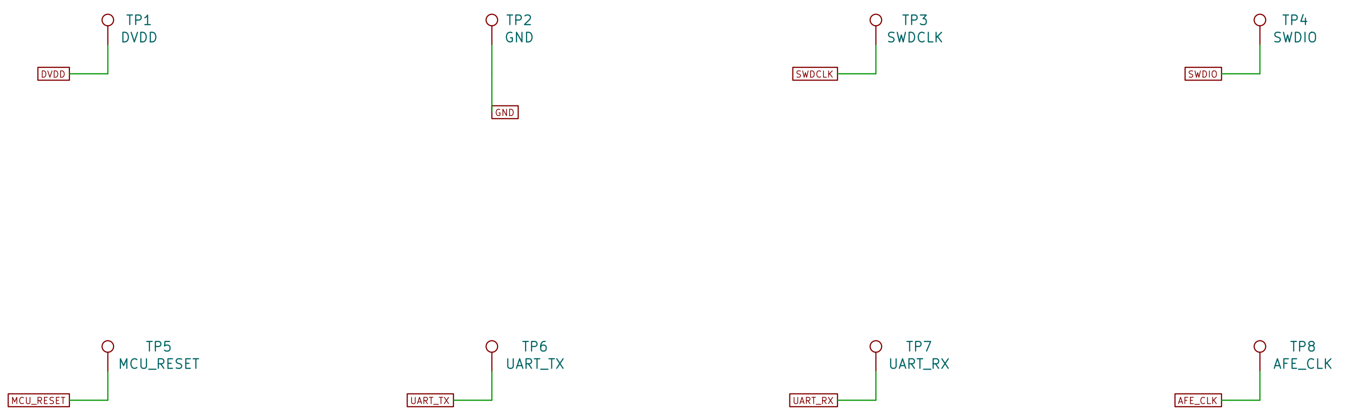
Battery voltage monitor



Status LED



Programming and test contacts



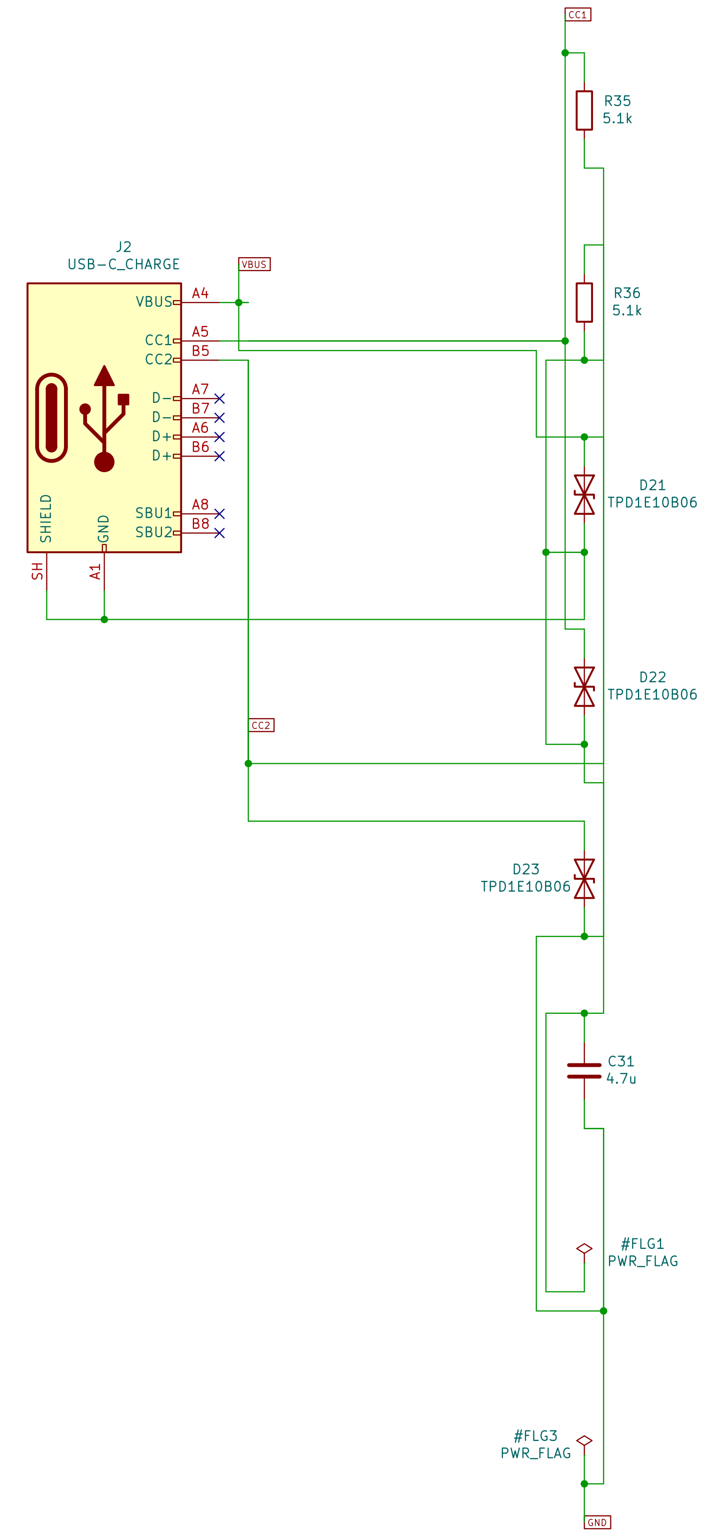
USB–C CHARGING / POWER PATH / DUAL LOW–NOISE LDO

100 mA USB input limit; 102.77mA nominal ISET programming, limited by input/load.

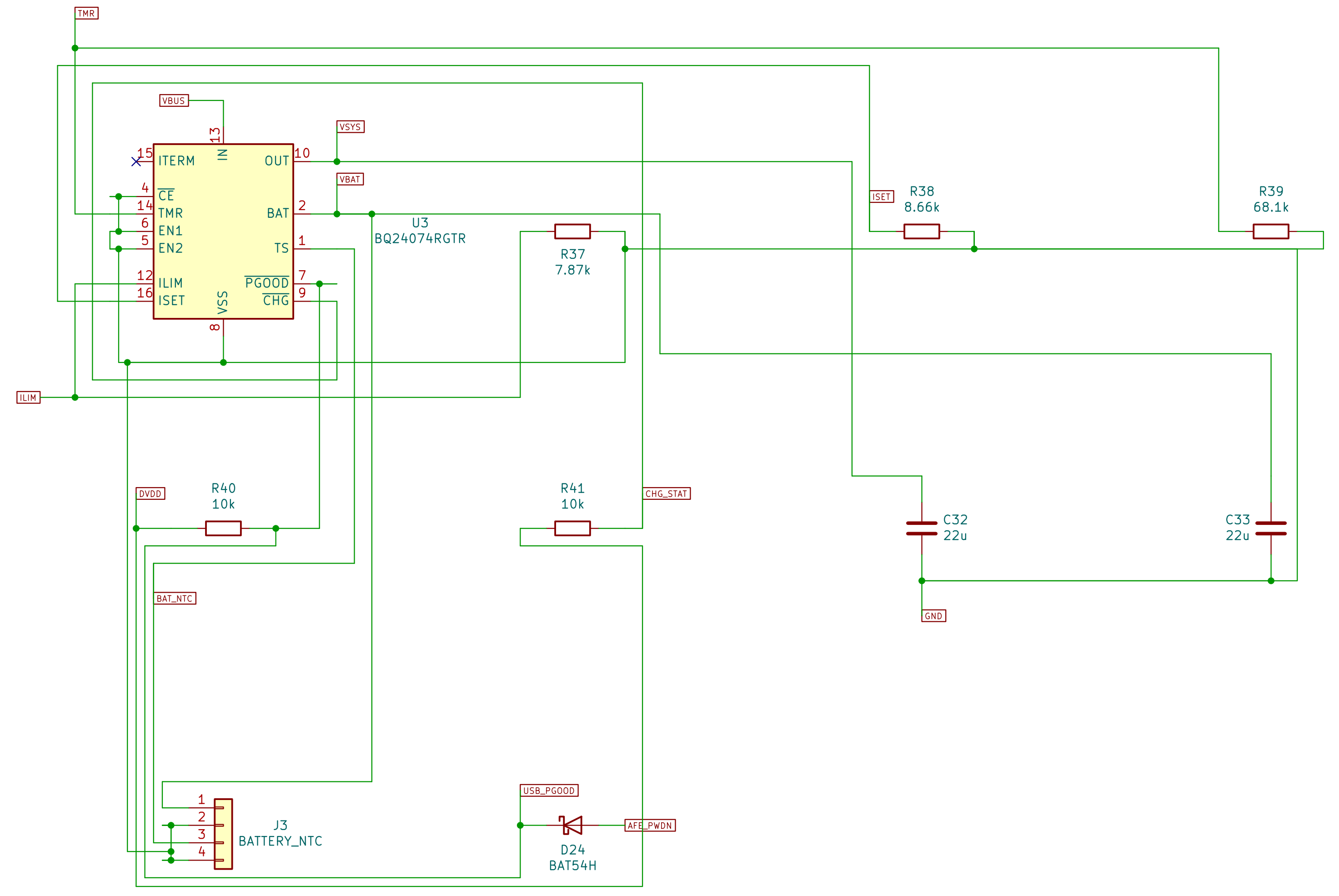
Protected 500 mAh 1S pack with thermally coupled 10k NTC is mandatory.

USB PGOOD diode forces AFE power–down. This is an acquisition interlock, not isolation.

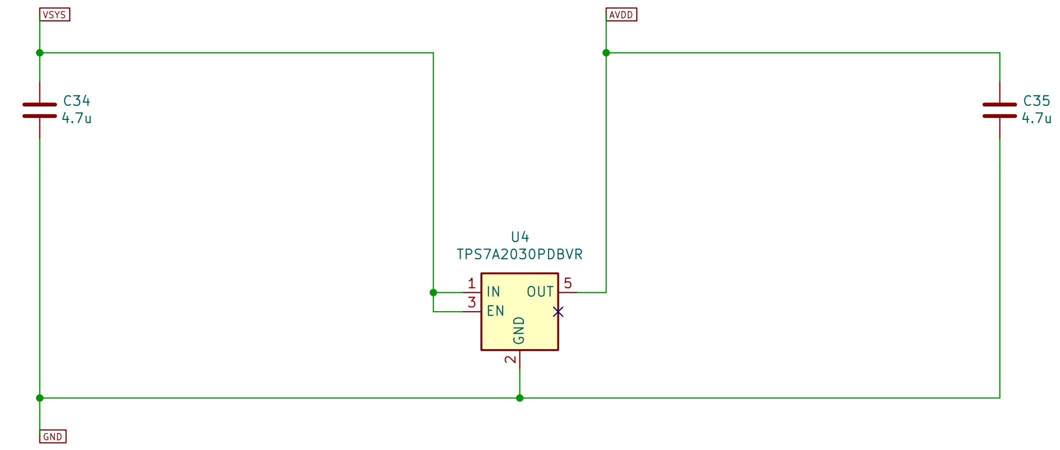
USB–C 5V input and CC protection



PowerPath charger and cell temperature



3V analog low–noise supply



3V digital supply